

PUSH DOWN AUTOMATA

- ✓ Definition of the Pushdown Automaton
- ✓ the Languages of a PDA
- ✓ Equivalence of PDA's and CFG's
- ✓ Acceptance by final state
- ✓ Acceptance by empty stack
- ✓ Deterministic Pushdown Automata.
- ✓ From CFG to PDA
- ✓ From PDA to CFG

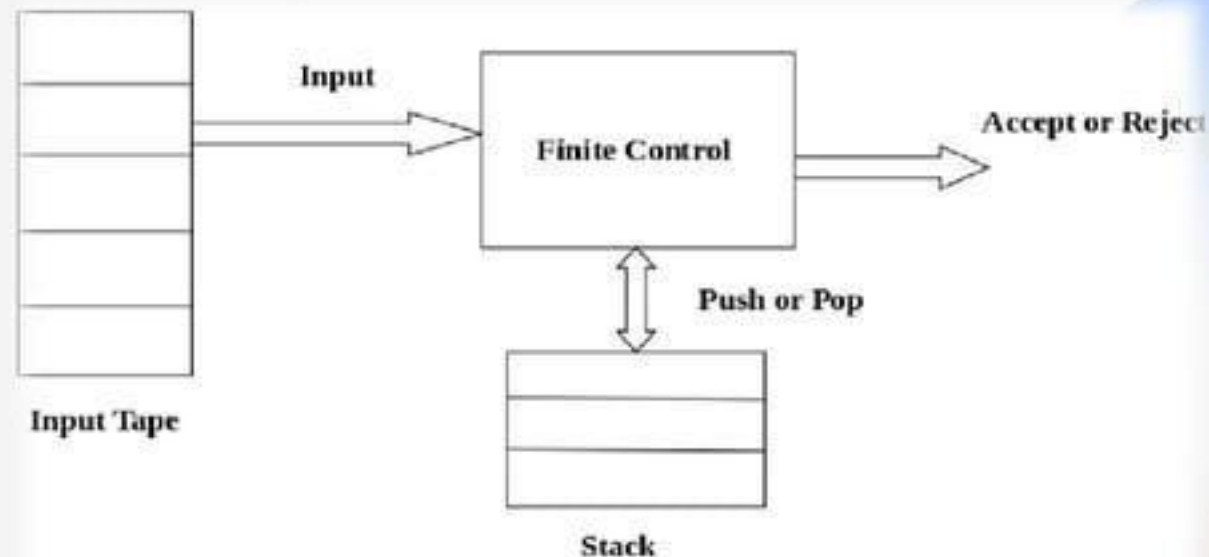


Introduction

- Pushdown automata is a way to implement a CFG in the same way we design DFA for a regular grammar. A DFA can remember a finite amount of information, but a PDA can remember an infinite amount of information.
- Pushdown automata is a way to implement a CFG in the same way we design DFA for a regular grammar.
- A DFA can remember a finite amount of information, but a PDA can remember an infinite amount of information.
- A PDA is more powerful than FA. Any language which can be acceptable by FA can also be acceptable by PDA. PDA also accepts a class of language which even cannot be accepted by FA. Thus PDA is much more superior to

Definition of PDA

Push Down automata is simply an NFA augmented with an “external stack memory”

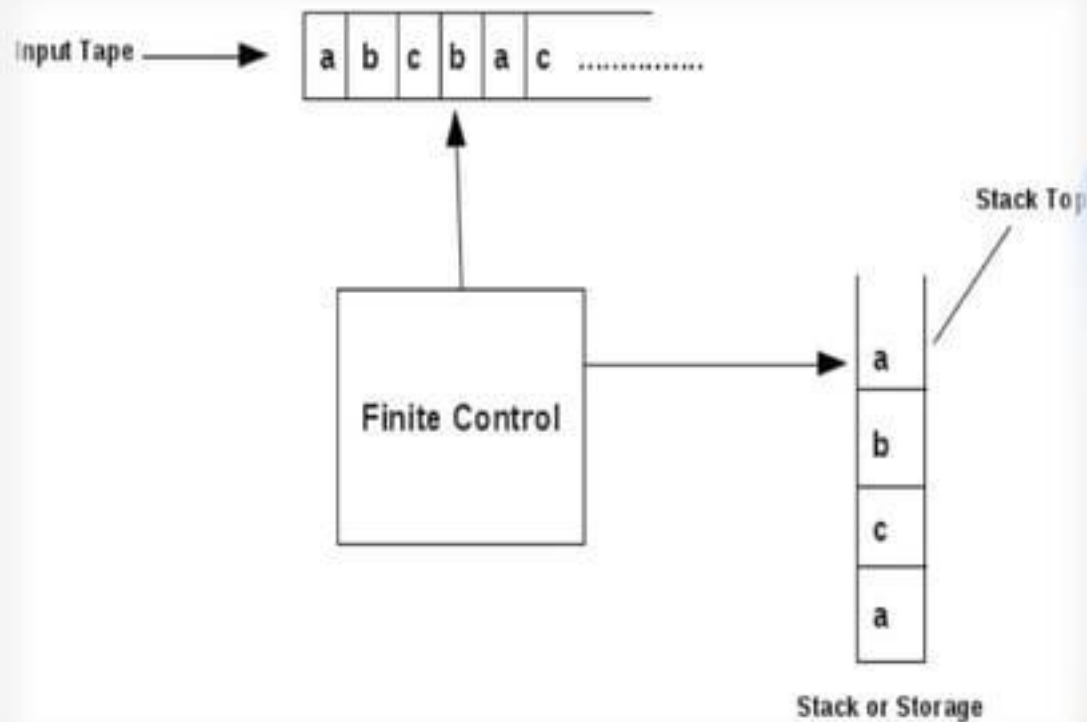


- PDA Components:**
1. Input tape
 2. Finite control
 3. Stack

Fig: Pushdown Automata

A PDA accepts the class of **Context-Free Languages (CFLs)**

PDA Components



Input tape: The input tape is divided in many cells or symbols. The input head is read-only and may only move from left to right, one symbol at a time.

Finite control: The finite control has some pointer which points the current symbol which is to be read.

Stack: The stack is a structure in which we can push and remove the items from one end only. It has an infinite size. In PDA, the stack is used to store the items temporarily.

Languages of a PDA

6 Definition of PDA

A pushdown automation, P is a system,

$$P = (Q, \Sigma, \Gamma, \delta, q_0, F)$$

where

Q is a set of states,

Σ is a set of input symbols,

Γ is a set of stack symbols (pushdown symbols),

q_0 is the start state,

F is a set of final states,

δ is a transition function which maps,

$$(Q \times \Sigma^* \times \Gamma^*) \longrightarrow (Q \times \Gamma^*)$$

Example

Consider a transition function, δ in a PDA defined by,

$$\delta = (p, a, \beta) \longrightarrow (q, \gamma)$$

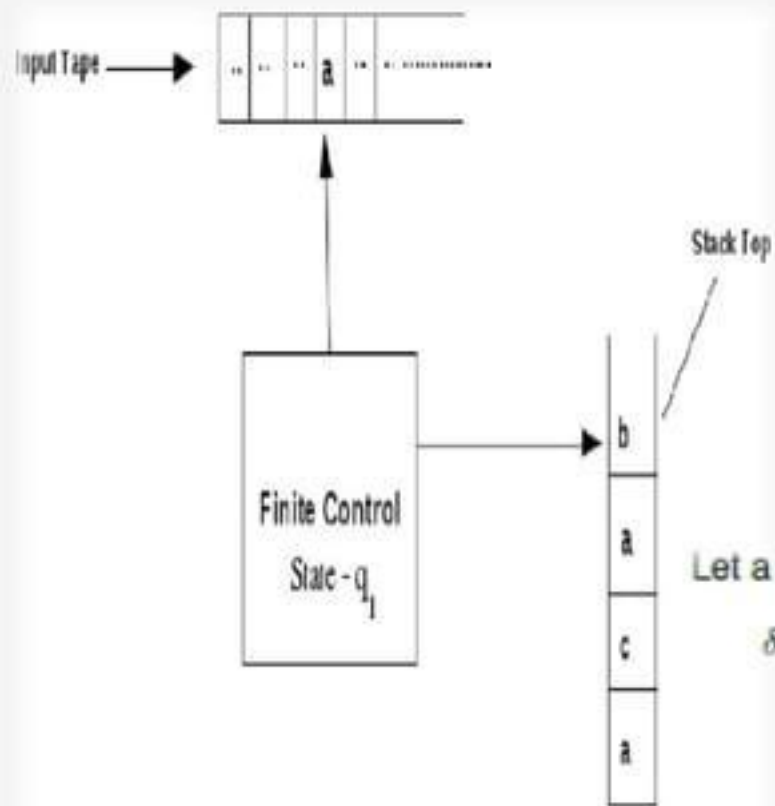
This means that

read a from the input tape,

pop the string β from stack top,

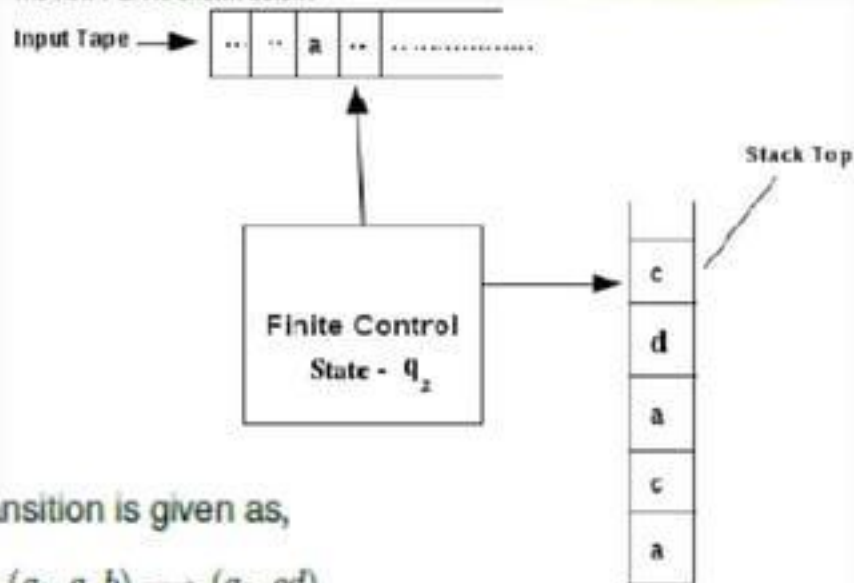
move from state p to state q ,

push string γ on to the stack top.



Stack or Storage

The new PDA is shown below:



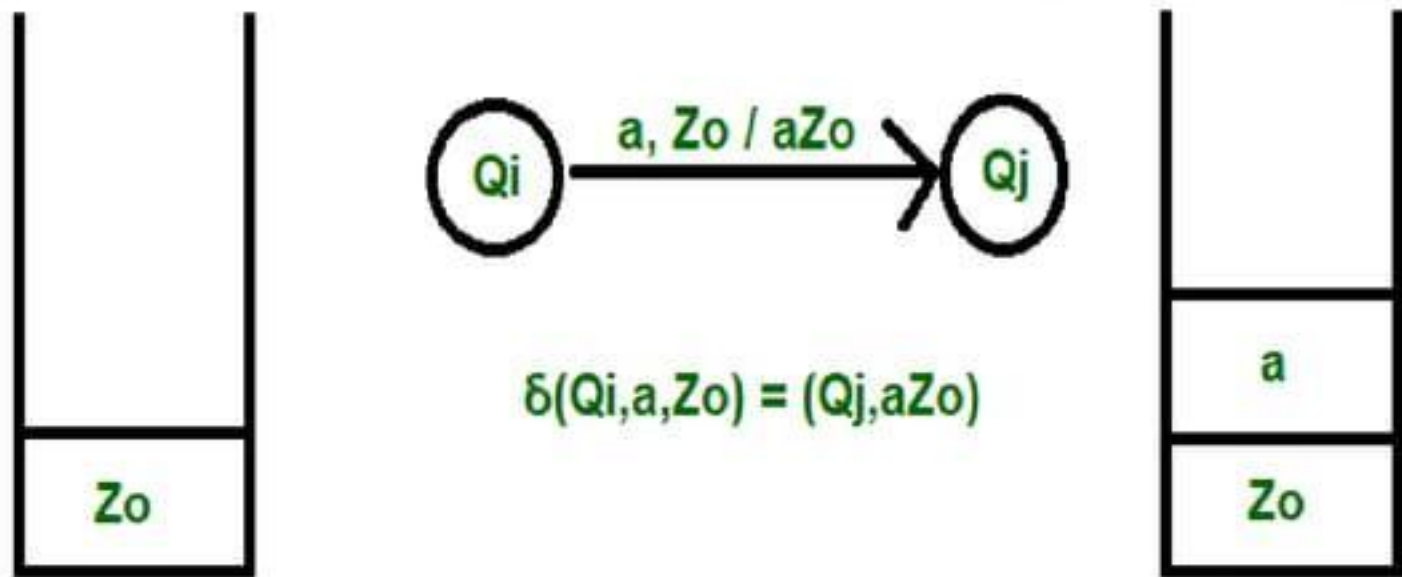
Stack or Storage

Let a transition is given as,

$$\delta = (q_1, a, b) \rightarrow (q_2, cd)$$

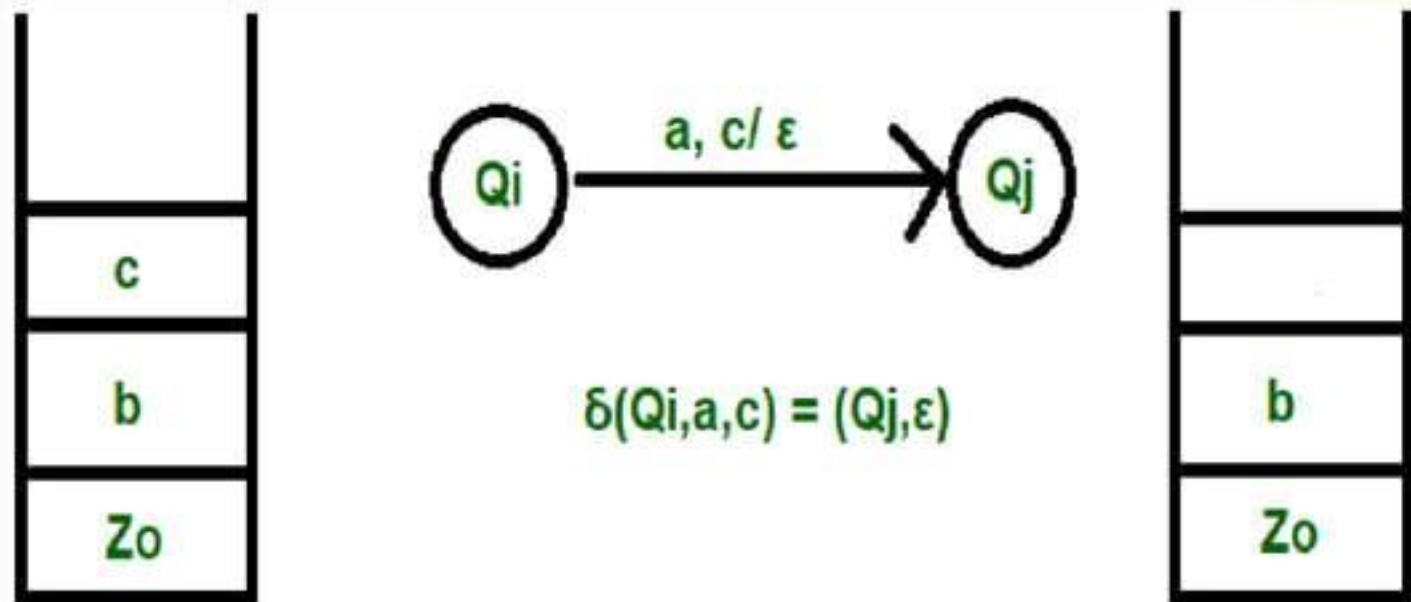
Representation of Stack in PDA

1. PUSH



The transition takes place in the order :- Input, Topmost Element / Final List Here, a is the input element, which is inserted into the stack, thus making the final content to be aZ_0 .

2.POP



The transition takes place in the order :- *Input Element, Topmost Element / Removal Confirmation* Here, a is the input, c is the element to be deleted and the removal confirmation is shown by Epsilon symbol declaring that the immediate has been popped and is Empty.

3.SKIP



$$\delta(Q_i, a, Z_0) = (Q_j, Z_0)$$



The transition takes place in the order:- *Input Element, Topmost element/Topmost Element* Here, a is the input and the stack remains unchanged after this operation.

Instantaneous Description (ID)

- ✓ ID is an informal notation of how a PDA computes an input string and make a decision that string is accepted or rejected.

An instantaneous description is a triple (q, w, α) where:

q describes the current state.

w describes the remaining input.

α describes the stack contents, top at the left.

Turnstile Notation:

$\vdash$ sign describes the turnstile notation and represents one move.

$\vdash^*$ sign describes a sequence of moves.

$$\delta(Q_i, a, Z_0) = (Q_j, aZ_0)$$

$$\delta(Q_i, a, c) = (Q_j, \epsilon)$$

Design a PDA for accepting a language $\{a^n b^n \mid n \geq 1\}$.

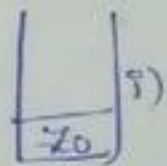
If $n=2$ then $a^2 b^2 \in aabb$ **Logic:** equal number of a's followed by equal number of b's.

First we will push all a's into the stack then reading every single b each a is popped from the stack.

Instantaneous Description for aa bb

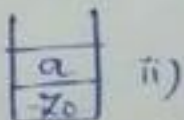
i) $\delta(q_0, \epsilon, \epsilon) = (q_0, z_0)$

current state current input top of the stack present state indicates empty stack.

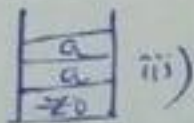


ii) $\delta(q_0, a, z_0) = (q_1, a z_0)$

pushing



iii) $\delta(q_1, a, a z_0) = (q_1, a a z_0)$

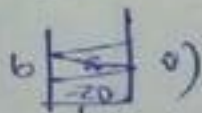


iv) $\delta(q_1, b, a) = (q_2, \epsilon)$

v) $\delta(q_2, b, a) = (q_2, \epsilon)$

vi) $\delta(q_2, \epsilon, z_0) = (q_3, \epsilon)$

pulling



Simulation

$$(q_0, \underline{a} a b b, \bar{z}_0) \vdash (q_0, \underline{a} b b, a \cdot \bar{z}_0)$$

$$\vdash (q_1, \underline{b} b, \cancel{a} \bar{z}_0)$$

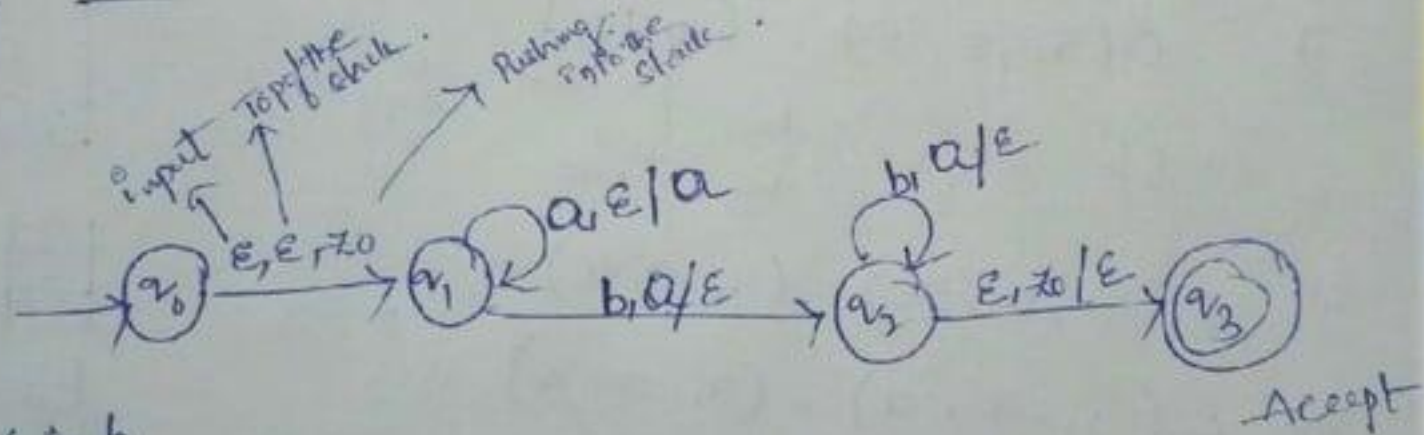
$$\vdash (q_2, \cancel{b}, \cancel{a} \bar{z}_0)$$

$$\vdash (q_2, \epsilon, \bar{z}_0)$$

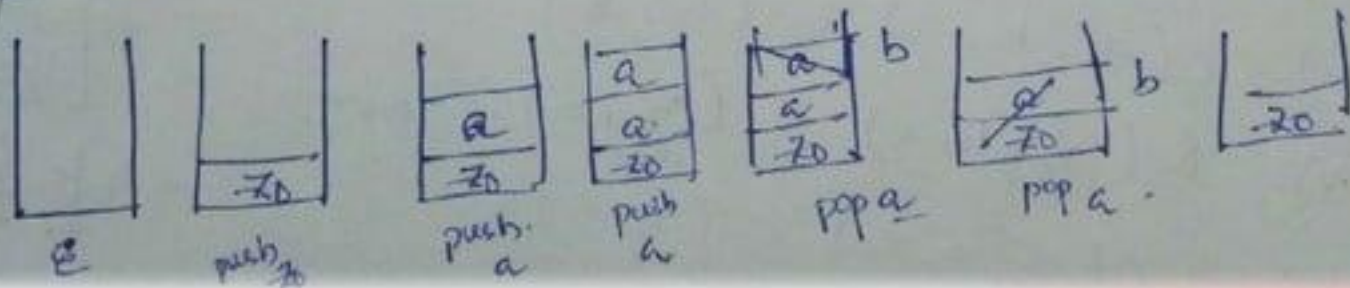
$$\vdash (q_3, \epsilon)$$

Accept state.

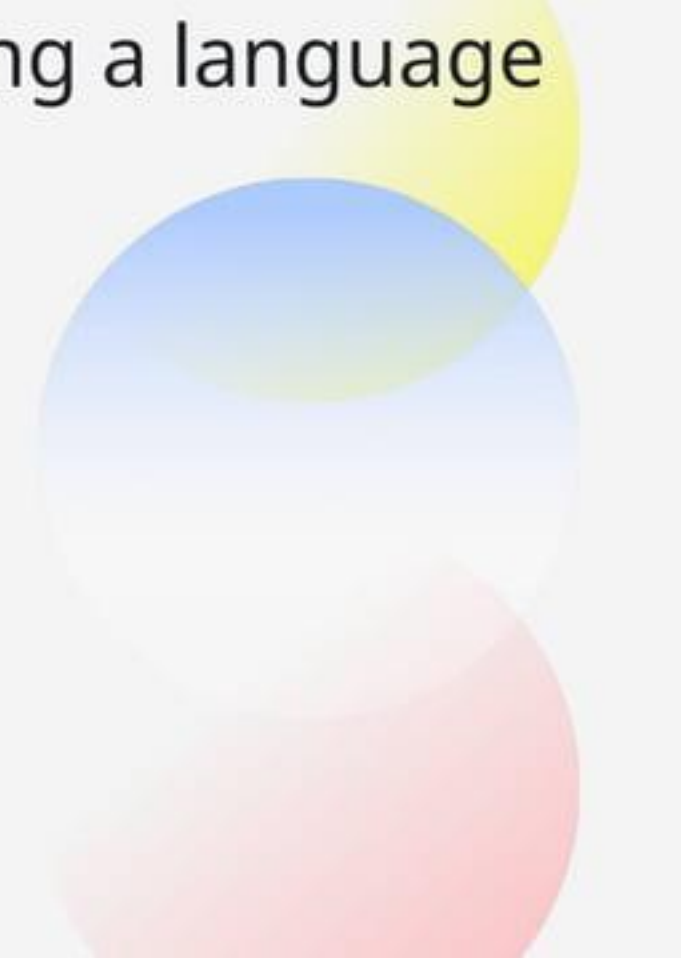
Graphical Notation for string aa bb



a b b



Design a PDA for accepting a language $\{a^n b^{2n} \mid n \geq 1\}$.



Consider a PDA,

$$P = (Q, \Sigma, \Gamma, \delta, q_0, F)$$

where

$$Q = \{s, f\}$$

$$\Sigma = \{a, b, c\}$$

$$\Gamma = \{a, b\}$$

$$q_0 = \{s\}$$

$$F = \{f\}$$

δ is given as follows:

$$1. (s, a, \varepsilon) \longrightarrow (s, a)$$

$$2. (s, b, \varepsilon) \longrightarrow (s, b)$$

$$3. (s, c, \varepsilon) \longrightarrow (f, \varepsilon)$$

$$4. (f, a, a) \longrightarrow (f, \varepsilon)$$

$$5. (f, b, b) \longrightarrow (f, \varepsilon)$$

Check whether the string *abacaba* is accepted by the above pushdown automation.

Acceptance in PDA

1. Acceptance by Final State

PDA **accepts** the input if, after reading the entire input string, it reaches a **final (accepting) state**. **Stack content** does **not matter** in this method.

Condition:

- ✓ Input is fully read.
- ✓ PDA is in a final/accepting state.

Example: Suppose PDA has state q_{accept} as final. If after processing the input, it reaches q_{accept} , the input is accepted.

2. Acceptance by Empty Stack

PDA accepts the input if, after reading the entire input, its stack becomes empty. Final state does not matter in this method.

Condition:

- ✓ Input is fully read.
- ✓ Stack is completely empty.

Example:

If after processing, the stack has no symbols left, input is accepted.

Equivalence of PDA's and CFG

1. CFG to PDA Conversion

2. PDA to CFG Conversion

1. CFG to PDA Conversion

The first symbol on R.H.S. production must be a terminal symbol.
The following steps are used to obtain PDA from CFG is:

Step 1: Convert the given productions of CFG into GNF.

Step 2: The PDA will only have one state $\{q\}$.

Step 3: The initial symbol of CFG will be the initial symbol in the PDA.

Step 4: For non-terminal symbol, add the following rule:

$$\delta(q, \epsilon, A) = (q, \alpha)$$

Where the production rule is $A \rightarrow \alpha$

Step 5: For each terminal symbols, add the following rule:

$$\delta(q, a, a) = (q, \epsilon) \text{ for every terminal symbol}$$

Convert the following grammar to a PDA that accepts the same language.

$$S \rightarrow 0SX \mid 1SY \mid \epsilon$$

$$X \rightarrow 1$$

$$Y \rightarrow 0$$

The PDA can be:

$$\mathbf{R1:} \delta(q, \epsilon, S) = \{(q, 0SX) \mid (q, 1SY) \mid (q, \epsilon)\}$$

$$\mathbf{R2:} \delta(q, \epsilon, X) = \{(q, 1)\}$$

$$\mathbf{R3:} \delta(q, \epsilon, Y) = \{(q, 0)\}$$

$$\mathbf{R4:} \delta(q, 0, 0) = \{(q, \epsilon)\}$$

$$\mathbf{R5:} \delta(q, 1, 1) = \{(q, \epsilon)\}$$

Construct PDA for the given CFG, and test whether 010^4 is acceptable by this PDA.

1. $S \rightarrow 0BB$

2. $B \rightarrow 0S \mid 1S \mid 0$

Solution:

The PDA can be given as:

3. $A = \{(q), (0, 1), (S, B, 0, 1), \delta, q, S, ?\}$

The production rule δ can be:

R1: $\delta(q, \varepsilon, S) = \{(q, 0BB)\}$

R2: $\delta(q, \varepsilon, B) = \{(q, 0S) \mid (q, 1S) \mid (q, 0)\}$

R3: $\delta(q, 0, 0) = \{(q, \varepsilon)\}$

R4: $\delta(q, 1, 1) = \{(q, \varepsilon)\}$

1
3
R2
R3

$\vdash \delta(q, 10000, BB)$	R1
$\vdash \delta(q, 10000, 1SB)$	R3
$\vdash \delta(q, 0000, SB)$	R2
$\vdash \delta(q, 0000, 0BBB)$	R
$\vdash \delta(q, 000, BBB)$	R
$\vdash \delta(q, 000, 0BB)$	
$\vdash \delta(q, 00, BB)$	
$\vdash \delta(q, 00, 0B)$	R2
$\vdash \delta(q, 0, B)$	R3
$\vdash \delta(q, 0, 0)$	R2
$\vdash \delta(q, \varepsilon)$	R3
ACCEPT	

Thus 010^4 is accepted by the PDA.

Draw a PDA for the CFG given below:

$S \rightarrow aSb$

$S \rightarrow a \mid b \mid \epsilon$

Solution:

The PDA can be given as:

1. $P = \{(q), (a, b), (S, a, b, z_0), \delta, q, z_0, q\}$ ^{R2}
R3

} The mapping function δ will be:

R1: $\delta(q, \epsilon, S) = \{(q, aSb)\}$

R2: $\delta(q, \epsilon, S) = \{(q, a) \mid (q, b) \mid (q, \epsilon)\}$

R3: $\delta(q, a, a) = \{(q, \epsilon)\}$

R4: $\delta(q, b, b) = \{(q, \epsilon)\}$

R5: $\delta(q, \epsilon, z_0) = \{(q, \epsilon)\}$

Simulation: Consider the string aaabb

$\delta(q, \epsilon aaabb, S) \vdash \delta(q, aaabb, aSb)$ R3

$\vdash \delta(q, \epsilon aabb, Sb)$ R1

$\vdash \delta(q, aabb, aSbb)$ R3

$\vdash \delta(q, \epsilon abb, Sbb)$

$\vdash \delta(q, abb, abb)$

$\vdash \delta(q, bb, bb)$ R4

$\vdash \delta(q, b, b)$ R4

$\vdash \delta(q, \epsilon, z_0)$ R5

$\vdash \delta(q, \epsilon)$

ACCEPT

2. PDA to CFG Conversion

Input – A CFG, $G = (V, T, P, S)$

Output – Equivalent PDA, $P = (Q, \Sigma, S, \delta, q_0, I, F)$

Step 1 – Convert the productions of the CFG into GNF.

Step 2 – The PDA will have only one state $\{q\}$.

Step 3 – The start symbol of CFG will be the start symbol in the PDA.

Step 4 – All non-terminals of the CFG will be the stack symbols of the PDA and all the terminals of the CFG will be the input symbols of the PDA.

Step 5 – For each production in the form $A \rightarrow aX$ where a is terminal and A, X are combination of terminal and non-terminals, make a transition $\delta(q, a, A)$.

Problem ① The PDA is given below

$$A = (\{q_0, q_1\}, \{0, 1\}, \{s, \Lambda\}, \delta, q_0, \$, \emptyset)$$

where δ is as given below.

$$\delta(q_0, 1, s) = \{(q_0, \Lambda s)\}$$

$$\delta(q_0, \epsilon, s) = \{(q_0, \epsilon)\}$$

$$\delta(q_0, 1, \Lambda) = \{(q_0, \Lambda 1)\}$$

$$\delta(q_0, 0, \Lambda) = \{(q_1, \Lambda)\}$$

$$\delta(q_1, 1, \Lambda) = \{(q_1, \epsilon)\}$$

$$\delta(q_1, 0, s) = \{(q_0, s)\}$$

Construct the CFG equivalent
to this PDA.

Sol

① $\delta(q_0, 1, S) = \{(q_0, A, S)\}$

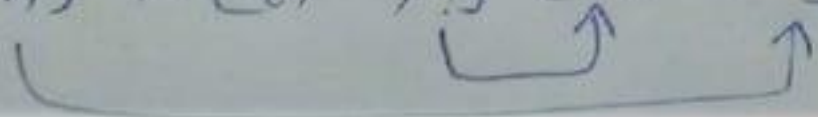
$\frac{R+1-S}{2} = \frac{2+1-5}{2} = \frac{-2}{2} = -1$
 $\frac{R+1-S}{2} = 4$
Trans

$[q_0, S, q_0] \rightarrow 1 [q_0, A, q_0] [q_0, S, q_0]$

$[q_0, S, q_0] \rightarrow 1 [q_0, A, q_1] [q_1, S, q_0]$

$[q_0, S, q_1] \rightarrow 1 [q_0, A, q_0] [q_0, S, q_1]$

$[q_0, S, q_1] \rightarrow 1 [q_0, A, q_1] [q_1, S, q_1]$



$$q_0, \epsilon, s) = (q_0, \epsilon)$$

$$[q_0, s, q_0] \rightarrow \epsilon$$

$$\begin{array}{r} \text{R.H.S} \quad \sum_{i=2}^{\infty} 0 \\ (q_0, \epsilon) \quad \sum_{i=2}^{\infty} 0 \\ \hline 1 \end{array}$$

$$\delta(q_0, 1, A) = (q_0, \underline{AA})$$

R.H.S

$$(q_0, \underline{AA})$$

$$2^2 = 4$$

transitions

$$[q_0, A, q_0] \rightarrow 1 [q_0, A, q_0] [q_0, A, q_0]$$

$$[q_0, A, q_0] \rightarrow 1 [q_0, A, q_1] [q_1, A, q_0]$$

$$[q_0, A, q_1] \rightarrow 1 [q_0, A, q_0] [q_0, A, q_1]$$

$$[q_0, A, q_1] \rightarrow 1 [q_0, A, q_1] [q_1, A, q_1]$$

$$\delta(q_0, 0, A) = (q_1, A)$$

$$[q_0, A, q_0] \rightarrow 0 [q_1, A, q_0]$$

$$[q_0, A, q_1] \rightarrow 0 [q_1, A, q_1]$$

$$R.H.S (q_1, A)_{2^1=2}$$

Correct

$$\delta(q_1, 1, A) = (q_1, \epsilon)$$

$$[q_1, A, q_0] \rightarrow 1 \{$$

$$R.H.S (q_1, \epsilon)_{2^1=1}$$

$$\delta(q_i, 0, S) = (q_0, S)$$

$$[q_i, S, q_0] \rightarrow O[q_0, S, q_0]$$

$$[q_i, S, q_i] \rightarrow O[q_0, S, q_i]$$

$$R.H.S. \left(\underline{q_0, S} \right)_1 \underset{2}{=} 2$$